Interventions against ash dieback in Great Britain

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Overview

Which interventions would best protect ash in Great Britain, and what decides the answer? This vignette runs the French model of vignette 5 on Great Britain with a changing host, to compare vaccinating trees, planting tolerant stock, felling and clearing leaf litter, to 2075 and on to 2100.

In short, which intervention comes out ahead depends mainly on how good the product is, when it can start and at what rate, and how far ahead you look. Long-term regeneration and the share of trees already symptomatic can also change the order. The fitted disease parameters and warming to 2075 change vaccine and planting results far less. The numbers are in the summary at the end, with the field data that would help most.

This is a comparison under assumptions, not a forecast. No ash vaccine exists, so vaccination is a generic strategy whose efficacy, duration and reach are inputs. The traits of tolerant stock

and the rate of regeneration are assumed too. Nothing is fitted to British data: the spread and symptom parameters come from France.

Every result below, in tables, figures and text, is computed by the code in this document, apart from two counts quoted from Forest Research's map of first finds, whose full data this package doesn't hold (see `data/gb/README.md`). The sweeps need a few thousand runs of a model taking several seconds each, so each saves its results to `results/` as it goes, and picks up from there. Delete a file there to recompute that sweep, or all of them to recompute everything, which takes hours.

Setup

```
using AshDieback
using CSV
using DataFrames
using LinearAlgebra
using Plots
using Printf
using Statistics

BLAS.set_num_threads(1)  # the threads go to the runs of a sweep
const RESULTS = joinpath(@__DIR__, "results")
mkpath(RESULTS)

"Print a Markdown table from a cell with `#| output: asis`."
function md_table(header, rows)
    println("| ", join(header, " | "), " |")
    println("|", repeat("---|", length(header)))
    foreach(r -> println("| ", join(r, " | "), " |"), rows)
    println()
end
fmt(x, d = 3) = @sprintf("%.*f", d, x)
interval(v) = (quantile(v, 0.025), quantile(v, 0.975))
fmti(v) = string(fmt(mean(v)), " [", fmt(interval(v)[1]), ", ",
 ↪  fmt(interval(v)[2]), "]")

const YELLOW, BLUE, ORANGE, AQUA, VIOLET = "#eda100", "#2a78d6", "#eb6834",
 ↪  "#1baf7a", "#6250d6"
nothing
```

Inputs

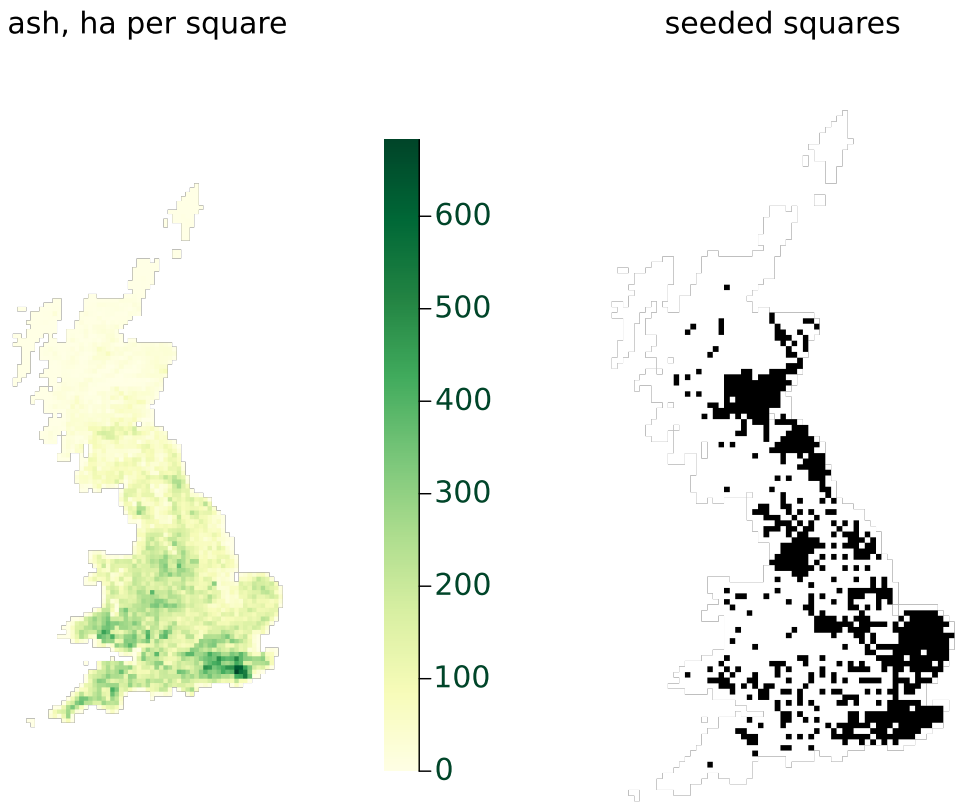
Great Britain is 10 km squares of the OS National Grid. Ash area comes from a modelled 1 km map. The epidemic starts in 2014 in the squares where it was first found in 2012–2015.

By October 2025, Forest Research's map shows it in 1779 squares, which hold 88 % of the ash (counted from the map and the ash data; the package holds only the seeded squares):

```
inputs = load_gb_inputs()
prob0, cell = gb_problem(inputs; B = 15.0)
K = prob0.data.ash_density
(; squares = length(inputs.squares), ash_ha = round(Int,
 ↪  sum(inputs.ash_ha)), squares_seeded = count(inputs.seeded),
 share_of_ash_in_seeded_squares = round(sum(inputs.ash_ha[inputs.seeded]) /
 ↪  sum(inputs.ash_ha); digits = 2))
```

```
(squares = 2827, ash_ha = 284072, squares_seeded = 734,
 ↪  share_of_ash_in_seeded_squares = 0.36)
```

```
ash_map, seed_map = fill(NaN, size(K)), fill(NaN, size(K))
for (k, s) in enumerate(inputs.squares)
    i, j = cell(s...)
    ash_map[i, j] = inputs.ash_ha[k]
    seed_map[i, j] = inputs.seeded[k] ? 1.0 : 0.0
end
plot(heatmap(ash_map'; title = "ash, ha per square", aspect_ratio = 1, axis
 ↪  = false, ticks = false, c = :YlGn),
     heatmap(seed_map'; title = "seeded squares", aspect_ratio = 1, axis =
      ↪  false, ticks = false, c = :Greys, colorbar = false);
     layout = (1, 2), size = (800, 600))
```



Stand basal area B converts ash area to the model's density, and it isn't measured directly. The National Forest Inventory's volume of ash, with an assumed form height, gives B of about 11 to 17 on this map's area if woodland alone counts, and trees outside woodland would raise it. So B = 15 is central here, with 5 to 35 tried below.

The host model and the interventions

The spore model stays as in France: production, diffusion, the cap S and the temperature function. `HostProblem` adds a changing host: classes of tree, juveniles, death and recruitment. Tree classes have a relative susceptibility, and leaf litter is produced in proportion to live ash.

These settings are estimates, mostly unmeasured. A native tolerant share of 5 % has a tenth of the death rate. Symptomatic trees die at $\mu = 0.04$ a year, which fits Europe-wide survival records. New trees arrive at $\rho = 0.03$ a year, for which only short studies exist.

```
const START = 2027

function host_for(ρ, μ, sc)
    sus = HostClass(:susceptible; share = 0.95, mortality = μ)
    # s_t sets the native tolerant class as well as planted stock, so each
     ↪  scenario has a matching baseline.
```

```
    tol = HostClass(:tolerant; share = 0.05, susceptibility = sc.s_t,
 ↪  mortality = 0.1 * μ)
    classes = [sus, tol]
    if sc.vaccine !== nothing
        push!(classes, vaccinated_class(sus; efficacy = sc.vaccine.efficacy,
         ↪  inoculum = 1.0, mortality = μ))
    end
    return HostDynamics(classes; background_mortality = 0.005,
     ↪  juvenile_years = 15.0, juvenile_mortality_factor = 2.5,
                         juvenile_inoculum = 0.1, regeneration_rate = ρ,
 ↪  heritability = 0.5)
end

const SCENARIOS = Dict{String,NamedTuple}()
register!(name; vaccine = nothing, ivs = AbstractIntervention[], s_t = 0.2)
 ↪  = (SCENARIOS[name] = (; name, vaccine, ivs, s_t); name)
base_of(name) = (s = SCENARIOS[name].s_t; s == 0.2 ? "baseline" : "baseline
 ↪  s $s")

function vaccinate!(name; efficacy = 0.9, waning = 0.05, max_coverage = 0.3,
 ↪  rate = 0.1, start = START, also = AbstractIntervention[])
    iv = Vaccinate(; from = :susceptible, to = :vaccinated_susceptible,
 ↪  years = start:2100, rate, max_coverage, waning,
                   symptomatic = false, juveniles = true, cells = nothing)
    return register!(name; vaccine = (; efficacy), ivs =
     ↪  AbstractIntervention[iv, also...])
end
plant_iv(rate, start, class = :tolerant) = PlantStock(; class, years =
 ↪  start:2075, rate)
plant_name(s, rate, start) = "plant s $s, $(100rate) %/y from $start"
plant!(s, rate, start) = register!(plant_name(s, rate, start); ivs =
 ↪  AbstractIntervention[plant_iv(rate, start)], s_t = s)

register!("baseline")
register!("baseline s 0.05"; s_t = 0.05)
register!("baseline s 0.5"; s_t = 0.5)
for start in (2027, 2040)
    vaccinate!("central vaccine from $start"; start)
    vaccinate!("100 % vaccine from $start"; start, max_coverage = 1.0, rate
     ↪  = 0.5)
    vaccinate!("perfect vaccine from $start"; start, efficacy = 1.0, waning
     ↪  = 0.0, max_coverage = 1.0, rate = 1.0)
    register!("plant ordinary, 2.0 %/y from $start"; ivs =
     ↪  AbstractIntervention[plant_iv(0.02, start, :susceptible)])
    foreach(s -> plant!(s, 0.02, start), (0.05, 0.5))
```

```
    plant!(0.05, 0.005, start)
end
for start in (2027, 2035, 2040, 2050), rate in (0.005, 0.01, 0.02)
    plant!(0.2, rate, start)
end
vaccinate!("efficacy 0.5"; efficacy = 0.5)
vaccinate!("lasts 5 years"; waning = 0.2)
vaccinate!("lifelong"; waning = 0.0)
vaccinate!("coverage up to 60 %"; max_coverage = 0.6)
vaccinate!("central vaccine + planting"; also =
 ↪  AbstractIntervention[plant_iv(0.02, START)])
register!("fell 50 %"; ivs = AbstractIntervention[Fell(; years = START:2100,
 ↪  rate = 0.5)])
register!("clear 50 % of litter"; ivs = AbstractIntervention[ReduceLitter(;
 ↪  years = START:2100, effect = 0.5)])
register!("clear 95 % of litter"; ivs = AbstractIntervention[ReduceLitter(;
 ↪  years = START:2100, effect = 0.95)])

const CENTRAL_V, HUNDRED_V, PERFECT_V = "central vaccine from 2027", "100 %
 ↪  vaccine from 2027", "perfect vaccine from 2027"
const PLANT, PLANT_T, PLANT_W, PLANT_O = plant_name(0.2, 0.02, 2027),
 ↪  plant_name(0.05, 0.02, 2027), plant_name(0.5, 0.02, 2027),
                                        "plant ordinary, 2.0 %/y from 2027"
length(SCENARIOS)
```

```
37
```

The central vaccine has efficacy 0.9, lasts 20 years on average, and reaches 10 % of healthy trees a year up to 30 %. It protects only trees not yet symptomatic. Tolerant stock has relative susceptibility s (0.2 central, 0.05 “very tolerant”). Planting fills gaps left by dead ash, up to 2075.

The measure is healthy adult ash in a given year, as a share of today’s ash area. It counts adult trees only, so it delays the credit for planting by the juvenile period of 15 years. Live area would hide the differences, since planting refills it whatever the stock’s quality. A gain is against the baseline of the same setting and the same s.

This is the machinery for the sweeps. Each case is one run, labelled by strings, and each finished case is appended to the sweep’s file:

```
const ASH = K .> 0
const SOUTH = let m = falses(size(K))
    foreach(s -> s[2] < 300 && (m[cell(s...)...] = true), inputs.squares)
    m .& ASH
end
const NORTH = let m = falses(size(K))
    foreach(s -> s[2] >= 600 && (m[cell(s...)...] = true), inputs.squares)
```

```
    m .& ASH
end
const OUTCOMES = [:year, :live, :adults, :healthy_adults, :q, :saturated,
 ↪  :mean_F, :treated_ha, :healthy_south, :healthy_north]

function outcome_rows(sol, years, B)
    K = sol.prob.prob.data.ash_density
    S = sol.prob.prob.p.S_sat
    ys = host_years(sol)
    live, adults = live_host(sol), live_host(sol; juveniles = false)
    q = symptom_share(sol)
    total = sum(K[ASH])
    wt = K[ASH] ./ total
    healthy_in(l, use) = sum(sum(view(sol.healthy, :, :, k, l)[use]) for k
     ↪  in axes(sol.healthy, 3))
    return map(years) do y
        l = y - first(ys) + 1
        F = view(sol.fungi, :, :, l - 1)[ASH]
        (; year = y, live = sum(view(live, :, :, l)[ASH]) / total, adults =
 ↪  sum(view(adults, :, :, l)[ASH]) / total,
         healthy_adults = healthy_in(l, ASH) / total, q = sum(wt .* view(q,
 ↪  :, :, l)[ASH]), saturated = sum(wt[F .>= 0.9999 * S]),
         mean_F = sum(wt .* F), treated_ha = isempty(sol.applied) ? 0.0 :
 ↪  sum(view(sol.applied, :, 1:l)) * 1e4 / B,
         healthy_south = healthy_in(l, SOUTH) / sum(K[SOUTH]), healthy_north
 ↪  = healthy_in(l, NORTH) / sum(K[NORTH]))
    end
end

function run_case(c)
    settings = ModelSettings(; initialization_year = c.init, final_year =
 ↪  maximum(c.years), delta_t = 1.0)
    sc = SCENARIOS[c.scenario]
    prob = AshDiebackProblem(c.data, c.θ; settings)
    return outcome_rows(solve(HostProblem(prob, host_for(c.ρ, c.μ, sc);
     ↪  interventions = sc.ivs)), c.years, c.B)
end

const THETA = [1000.0, 0.0, 23.26, 0.0, 18.51, 92.0, 21.50001, 1.0, 0.05,
 ↪  0.056]
const DATA15 = prob0.data
const DEFAULT_YEARS = (2026, 2050, 2075)
case(labels, scenario; data = DATA15, B = 15.0, θ = THETA, ρ = 0.03, μ =
 ↪  0.04, years = DEFAULT_YEARS, init = 2014) =
    (; labels, scenario, data, B, θ, ρ, μ, years, init)
```

```
"Run the cases not yet in results/name.csv, then return the rows of all the
 ↪  cases."
function sweep(name, cases)
    path = joinpath(RESULTS, name * ".csv")
    labels = collect(keys(first(cases).labels))
    key_columns = vcat(labels, :scenario)
    case_key(c) = join((values(c.labels)..., c.scenario), "|")
    row_key(r) = join((r[l] for l in key_columns), "|")
    wanted = Dict(case_key(c) => length(c.years) for c in cases)
    @assert length(wanted) == length(cases)
    read_rows() = unique(CSV.read(path, DataFrame; types = Dict(l => String
     ↪  for l in key_columns), stringtype = String),
                         vcat(key_columns, :year); keep = :last)
    finished = Set{String}()
    if isfile(path)
        for g in groupby(read_rows(), key_columns)
            k = row_key(first(g))
            get(wanted, k, -1) == nrow(g) && push!(finished, k)
        end
    end
    todo = filter(c -> case_key(c) ∉ finished, cases)
    if !isempty(todo)
        isfile(path) || write(path, join(vcat(key_columns, OUTCOMES), ",") *
         ↪  "\n")
        lk = ReentrantLock()
        io = open(path, "a")
        Threads.@threads :dynamic for n in eachindex(todo)
            c = todo[n]
            out = run_case(c)
            text = join((join((values(c.labels)..., "\"" * c.scenario *
 ↪  "\"",
                               (r[f] isa Int ? r[f] : round(r[f]; sigdigits
 ↪  = 7) for f in OUTCOMES)...), ",") for r in out), "\n") * "\n"
            lock(lk) do
                write(io, text)
                flush(io)
            end
        end
        close(io)
    end
    selected = filter(r -> haskey(wanted, row_key(r)), read_rows())
    @assert nrow(selected) == sum(values(wanted)) "$name: expected
     ↪  $(sum(values(wanted))) rows, found $(nrow(selected))"
    return selected
end
```

```
"Lookups into a sweep: `value(labels..., scenario)` and `gain(labels...,
 ↪  scenario)`, for one year and column."
function lookups(df; year = 2075, col = :healthy_adults)
    labels = [n for n in propertynames(df) if n ∉ OUTCOMES]
    d = filter(:year => ==(year), df)
    v = Dict(Tuple(String[string(r[l]) for l in labels]) => Float64(r[col])
 ↪  for r in eachrow(d))
    value(args...) = v[Tuple(String[string(a) for a in args])]
    gain(args...) = value(args...) - value(args[1:(end - 1)]...,
     ↪  base_of(args[end]))
    return (; value, gain)
end
nothing
```

The ranking at each horizon

The central case, at the French best-fit parameters, year by year to 2100:

```
central_names = ["baseline", "baseline s 0.05", "baseline s 0.5",
 ↪  sort(filter(n -> !startswith(n, "baseline"),
 ↪  collect(keys(SCENARIOS))))...]
timing = sweep("timing", [case((; setting = "central"), n; years =
 ↪  2026:2100) for n in central_names])
nrow(timing)
```

```
2775
```

```
gain_y(name, y; col = :healthy_adults) = lookups(timing; year = y,
 ↪  col).gain("central", name)
value_y(name, y; col = :healthy_adults) = lookups(timing; year = y,
 ↪  col).value("central", name)
"Mean area treated a year over 2027–2075, ha, to two significant figures."
ha_per_year(name) = round(Int, round(value_y(name, 2075; col = :treated_ha)
 ↪  / 49; sigdigits = 2))
peak_year(name) = (2026:2100)[argmax([gain_y(name, y) for y in 2026:2100])]
base75 = (healthy = value_y("baseline", 2075), adults = value_y("baseline",
 ↪  2075; col = :adults),
          q26 = value_y("baseline", 2026; col = :q), q75 =
 ↪  value_y("baseline", 2075; col = :q),
          sat75 = value_y("baseline", 2075; col = :saturated), live75 =
 ↪  value_y("baseline", 2075; col = :live))
map(x -> round(x; digits = 3), base75)
```

```
(healthy = 0.062, adults = 0.26, q26 = 0.527, q75 = 0.78, sat75 = 0.931,
 ↪  live75 = 0.322)
```

Without any intervention, healthy adults in 2075 are 0.06 of today's ash, and all adults 0.26. One point (0.01) is about 2841 ha. Each gain:

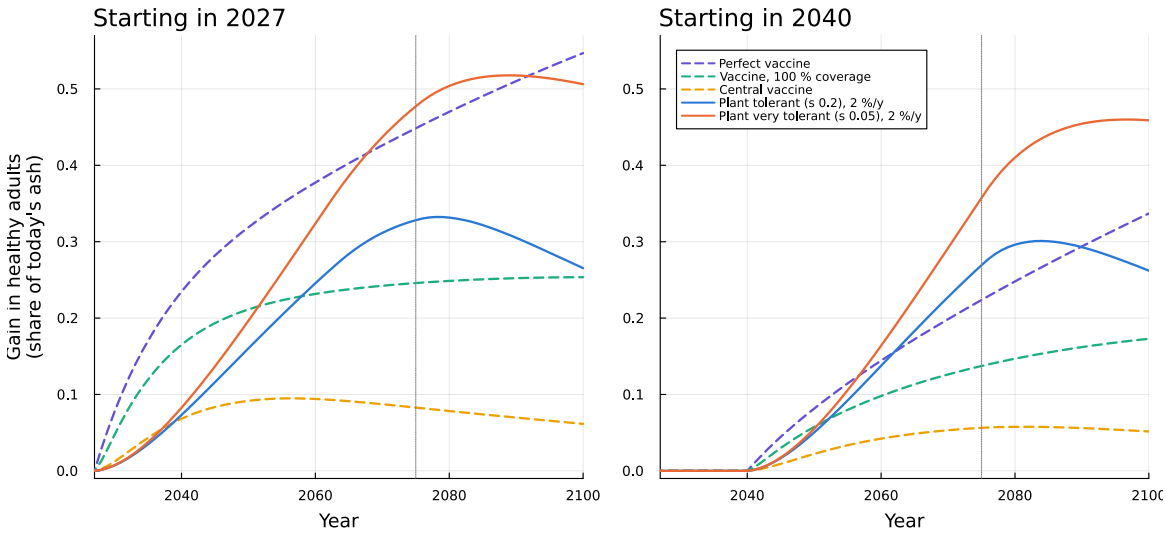
```
ranked = [PLANT_T, PERFECT_V, "clear 95 % of litter", "central vaccine +
 ↪  planting", PLANT, HUNDRED_V, PLANT_W, CENTRAL_V, PLANT_O,
          "fell 50 %", "clear 50 % of litter"]
md_table(["From 2027", "2050", "2075", "2100", "Change in all adults,
 ↪  2075"],
         [[n, fmt(gain_y(n, 2050)), "**" * fmt(gain_y(n, 2075)) * "**",
 ↪  fmt(gain_y(n, 2100)), @sprintf("%+.2f", gain_y(n, 2075; col = :adults))]
          for n in ranked])
```

From 2027	2050	2075	2100	Change in all adults, 2075
plant s 0.05, 2.0 %/y from 2027	0.197	0.477	0.506	+0.51
perfect vaccine from 2027	0.319	0.448	0.547	+0.33
clear 95 % of litter	0.252	0.400	0.515	+0.27
central vaccine + planting	0.248	0.364	0.279	+0.53
plant s 0.2, 2.0 %/y from 2027	0.160	0.328	0.265	+0.51
100 % vaccine from 2027	0.211	0.246	0.254	+0.20
plant s 0.5, 2.0 %/y from 2027	0.108	0.165	0.076	+0.50
central vaccine from 2027	0.092	0.083	0.062	+0.08
plant ordinary, 2.0 %/y from 2027	0.057	0.072	0.013	+0.25
fell 50 %	0.006	0.041	0.086	-0.16
clear 50 % of litter	0.006	0.015	0.032	+0.01

At 2 %/y from 2027, planting's gain peaks in 2078 at s 0.2 and 2089 at s 0.05, then falls, probably because these runs stop planting in 2075. At s 0.5 it already peaks in 2072, before the stop. So the 2100 column for planting shows what follows a stop, not a forecast with planting continuing.

Timing decides between vaccines and planting

```
series = [("Perfect vaccine", "perfect vaccine from ", VIOLET, :dash),
 ↪  ("Vaccine, 100 % coverage", "100 % vaccine from ", AQUA, :dash),
          ("Central vaccine", "central vaccine from ", YELLOW, :dash)]
wide = sort(unstack(timing, :year, :scenario, :healthy_adults), :year)
@assert wide.year == 2026:2100
curve(name) = wide[!, name] .- wide[!, base_of(name)]
function timing_panel(start; kw...)
    p = plot(; title = "Starting in $start", xlabel = "Year", ylims =
 ↪  (-0.01, 0.57), xlims = (2027, 2100), titlelocation = :left, kw...)
    for (label, prefix, colour, style) in series
        plot!(p, 2026:2100, curve(prefix * string(start)); label, color =
         ↪  colour, linestyle = style, linewidth = 2)
    end
    plot!(p, 2026:2100, curve(plant_name(0.2, 0.02, start)); label = "Plant
     ↪  tolerant (s 0.2), 2 %/y", color = BLUE, linewidth = 2)
    plot!(p, 2026:2100, curve(plant_name(0.05, 0.02, start)); label = "Plant
     ↪  very tolerant (s 0.05), 2 %/y", color = ORANGE, linewidth = 2)
    vline!(p, [2075]; color = :gray, linestyle = :dot, label = "")
    return p
end
plot(timing_panel(2027; ylabel = "Gain in healthy adults\n(share of today's
 ↪  ash)", legend = false),
     timing_panel(2040; legend = :topleft, legendfontsize = 7); layout = (1,
      ↪  2), size = (1000, 460), left_margin = 4Plots.mm, bottom_margin =
      ↪  4Plots.mm)
```



Dashed lines are vaccines, solid lines planting. French best-fit parameters, $B = 15$, no warming.

The curves cross, so the ranking depends on the horizon. Each entry is when the first overtakes the second, for good unless stated:

```
function crossings(a, b, y0, y1 = 2100)
    ca, cb = curve(a), curve(b)
    out = String[]
    previous = nothing
    for (i, y) in enumerate(2026:y1)
        y < y0 && continue
        ahead = ca[i] > cb[i]
        previous !== nothing && ahead != previous && push!(out, ahead ?
 ↪  string(y) : "behind again from $y")
        previous = ahead
    end
    return isempty(out) ? (previous ? "ahead from the start" : "not by $y1")
     ↪  : join(out, ", ")
end
md_table(["Crossover", "Both from 2027", "Both from 2040"],
         [["Tolerant planting (s 0.2, 2 %/y) passes central vaccine",
 ↪  crossings(PLANT, CENTRAL_V, 2029),
           crossings(plant_name(0.2, 0.02, 2040), "central vaccine from
            ↪  2040", 2042)],
          ["Tolerant planting (s 0.2, 2 %/y) passes 100 % vaccine",
 ↪  crossings(PLANT, HUNDRED_V, 2029),
           crossings(plant_name(0.2, 0.02, 2040), "100 % vaccine from 2040",
            ↪  2042)],
          ["Very tolerant planting (s 0.05, 2 %/y) passes perfect vaccine",
 ↪  crossings(PLANT_T, PERFECT_V, 2029),
           crossings(plant_name(0.05, 0.02, 2040), "perfect vaccine from
            ↪  2040", 2042)]])
```

Crossover	Both from 2027	Both from 2040
Tolerant planting (s 0.2, 2 %/y) passes central vaccine	2039	ahead from the start
Tolerant planting (s 0.2, 2 %/y) passes 100 % vaccine	2058	2053
Very tolerant planting (s 0.05, 2 %/y) passes perfect vaccine	2068, behind again from 2092	2057

A delay hits vaccines harder, as they protect only trees still healthy, whereas planting probably replaces trees that have died, and more have died by then. Starting in 2040 rather than 2027 costs a perfect vaccine 50 % of its 2075 gain, the 100 % vaccine 44 %, and tolerant planting at 2 %/y 18 %.

Supply matters as much as the start date. Every run so far starts planting at 2 %/y in 2027, which is about 5000 ha of ash a year, and 0.5 %/y is about 1400 ha. All new UK woodland was 15 580 ha in 2024/25, though replanting ash on existing sites wouldn't count in that figure. Tolerant ash is at the archive stage, so these rates from 2027 are probably not achievable. Gains in 2075:

```
md_table(["Tolerant planting (s 0.2)", "0.5 %/y", "1 %/y", "2 %/y"],
         [["From $s", (fmt(gain_y(plant_name(0.2, r, s), 2075)) for r in
 ↪  (0.005, 0.01, 0.02))...] for s in (2027, 2035, 2040, 2050)])
md_table(["Vaccine", "From 2027", "From 2040"],
         [[label, fmt(gain_y(p * "2027", 2075)), fmt(gain_y(p * "2040",
 ↪  2075))]
          for (label, p) in (("Central", "central vaccine from "), ("100 %
 ↪  coverage", "100 % vaccine from "), ("Perfect", "perfect vaccine from
 ↪  "))])
```

Tolerant planting (s 0.2)	0.5 %/y	1 %/y	2 %/y
From 2027	0.095	0.186	0.328
From 2035	0.080	0.158	0.305
From 2040	0.070	0.138	0.269
From 2050	0.047	0.094	0.186

Vaccine	From 2027	From 2040
Central	0.083	0.056
100 % coverage	0.246	0.137
Perfect	0.448	0.224

When tolerant planting at the lowest rate passes the central vaccine started in 2027:

```
first_pass(name) = crossings(name, CENTRAL_V, 2029, 2075)
(; from_2027 = first_pass(plant_name(0.2, 0.005, 2027)), from_2035 =
 ↪  first_pass(plant_name(0.2, 0.005, 2035)),
 from_2040 = first_pass(plant_name(0.2, 0.005, 2040)))
```

```
(from_2027 = "2071", from_2035 = "not by 2075", from_2040 = "not by 2075")
```

Product quality

Quality moves results as far as anything else here. A vaccine's efficacy and duration matter more than its coverage cap, because reaching 10 % a year is what limits it:

```
quality_v = [("Efficacy 0.5", "efficacy 0.5"), ("Lasts 5 years", "lasts 5
 ↪  years"), ("Central: efficacy 0.9, 20 years", CENTRAL_V),
            ("Coverage up to 60 %", "coverage up to 60 %"), ("Lifelong",
 ↪  "lifelong"), ("100 % coverage, reaching 50 % a year", HUNDRED_V),
            ("Perfect: all healthy trees at once, for life", PERFECT_V)]
md_table(["Vaccine, from 2027", "Gain in 2075"], [[a, fmt(gain_y(n, 2075))]
 ↪  for (a, n) in quality_v])
md_table(["Planting 2 %/y from 2027", "Gain in 2075"],
         [["Ordinary stock", fmt(gain_y(PLANT_O, 2075))], ["s 0.5",
 ↪  fmt(gain_y(PLANT_W, 2075))], ["s 0.2", fmt(gain_y(PLANT, 2075))],
          ["s 0.05", fmt(gain_y(PLANT_T, 2075))]])
```

Vaccine, from 2027	Gain in 2075
Efficacy 0.5	0.026
Lasts 5 years	0.031
Central: efficacy 0.9, 20 years	0.083
Coverage up to 60 %	0.093
Lifelong	0.103
100 % coverage, reaching 50 % a year	0.246
Perfect: all healthy trees at once, for life	0.448

Planting 2 %/y from 2027	Gain in 2075
Ordinary stock	0.072
s 0.5	0.165
s 0.2	0.328
s 0.05	0.477

Regeneration and tree death

Regeneration ρ and the death of symptomatic trees μ are the host model's main unknowns. Each run here is at $B = 25$:

```
data25 = gb_problem(inputs; B = 25.0)[1].data
const RHOS, MUS = (0.01, 0.03, 0.1), (0.02, 0.04, 0.1)
grid_names = ["baseline", "baseline s 0.05", "baseline s 0.5", CENTRAL_V,
 ↪  HUNDRED_V, PERFECT_V, PLANT, PLANT_T, PLANT_W, PLANT_O,
              plant_name(0.2, 0.005, 2027)]
grid_runs = sweep("regeneration", vec([case((; rho = string(ρ), mu =
 ↪  string(μ)), n; data = data25, B = 25.0, ρ, μ)
                                       for ρ in RHOS, μ in MUS, n in
 ↪  grid_names]))
```

```
gg = lookups(grid_runs).gain
corners = [(ρ, μ) for ρ in RHOS for μ in MUS]
grid_range(n) = extrema(gg(ρ, μ, n) for (ρ, μ) in corners)
beats(a, b) = [(ρ, μ) for (ρ, μ) in corners if gg(ρ, μ, a) > gg(ρ, μ, b)]
nrow(grid_runs)
```

```
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```

```
md_table(["Scenario", "Lowest gain over the grid", "Highest"],
         [[n, fmt(grid_range(n)[1]), fmt(grid_range(n)[2])] for n in
 ↪  (PLANT_T, PERFECT_V, PLANT, HUNDRED_V, PLANT_W, CENTRAL_V, PLANT_0)])
md_table(["Which beats which, of 9 cells (ρ, μ)", "Cells", "Where it
 ↪  loses"],
         [[string(a, " > ", b), length(beats(a, b)),
 ↪  join(string.(setdiff(corners, beats(a, b))), " ")]
          for (a, b) in ((PLANT, CENTRAL_V), (PLANT, HUNDRED_V), (PLANT_T,
 ↪  PERFECT_V), (PLANT_W, CENTRAL_V), (plant_name(0.2, 0.005, 2027),
 ↪  CENTRAL_V))])
```

Scenario	Lowest gain over the grid	Highest
plant s 0.05, 2.0 %/y from 2027	0.298	0.541
perfect vaccine from 2027	0.344	0.637
plant s 0.2, 2.0 %/y from 2027	0.189	0.377
100 % vaccine from 2027	0.175	0.388
plant s 0.5, 2.0 %/y from 2027	0.073	0.198
central vaccine from 2027	0.027	0.120
plant ordinary, 2.0 %/y from 2027	0.020	0.085

Which beats which, of 9 cells (ρ, μ)	Cells	Where it loses
plant s 0.2, 2.0 %/y from 2027 > central vaccine from 2027	9	
plant s 0.2, 2.0 %/y from 2027 > 100 % vaccine from 2027	5	(0.03, 0.02) (0.1, 0.02) (0.1, 0.04) (0.1, 0.1)
plant s 0.05, 2.0 %/y from 2027 > perfect vaccine from 2027	4	(0.01, 0.02) (0.03, 0.02) (0.1, 0.02) (0.1, 0.04) (0.1, 0.1)
plant s 0.5, 2.0 %/y from 2027 > central vaccine from 2027	7	(0.1, 0.02) (0.1, 0.04)
plant s 0.2, 0.5 %/y from 2027 > central vaccine from 2027	7	(0.1, 0.02) (0.1, 0.04)

So high regeneration can make even a 100 % vaccine beat s 0.2 planting. Long-term regeneration in affected stands is also the input with the least evidence.

The share already symptomatic

A vaccine helps only trees still healthy, so it depends on how many have symptoms by 2026, which the model gives but doesn't observe. Here the epidemic starts earlier or later, or the yearly rate of new symptoms changes (r_S doubled or halved, at B = 25):

```
share_settings = (("seeded 2010", 1.0, 2010), ("seeded 2014 (central)", 1.0,
 ↪  2014), ("seeded 2018", 1.0, 2018),
                  ("rate halved", 2.0, 2014), ("rate doubled", 0.5, 2014))
theta_rate(rs) = [1000.0 * rs, 0.0, 23.26, 0.0, 18.51, 92.0, 21.50001, 1.0,
 ↪  0.05, 0.056]
share_names = ["baseline", "baseline s 0.05", CENTRAL_V, HUNDRED_V,
 ↪  PERFECT_V, PLANT, PLANT_T]
share_runs = sweep("share_symptomatic", [case((; setting = l), n; data =
 ↪  data25, B = 25.0, θ = theta_rate(rs), init)
                                          for (l, rs, init) in share_settings
 ↪  for n in share_names])
sg, sq = lookups(share_runs).gain, lookups(share_runs; year = 2026, col =
 ↪  :q).value
nrow(share_runs)
```

```
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```

```
md_table(["Setting", "q in 2026", "Central vaccine", "100 % vaccine",
 ↪  "Perfect vaccine", "Plant s 0.2", "Plant s 0.05"],
         [[l, fmt(sq(l, "baseline"), 2), (fmt(sg(l, n)) for n in (CENTRAL_V,
 ↪  HUNDRED_V, PERFECT_V, PLANT, PLANT_T))...] for (l, _, _) in
 ↪  share_settings])
```

Setting	q in 2026	Central vaccine	100 % vaccine	Perfect vaccine	Plant s 0.2	Plant s 0.05
seeded 2010	0.63	0.073	0.210	0.381	0.339	0.491
seeded 2014 (central)	0.53	0.083	0.246	0.450	0.331	0.482
seeded 2018	0.39	0.094	0.297	0.548	0.322	0.470
rate halved	0.32	0.116	0.360	0.521	0.334	0.412
rate doubled	0.79	0.029	0.109	0.299	0.231	0.457

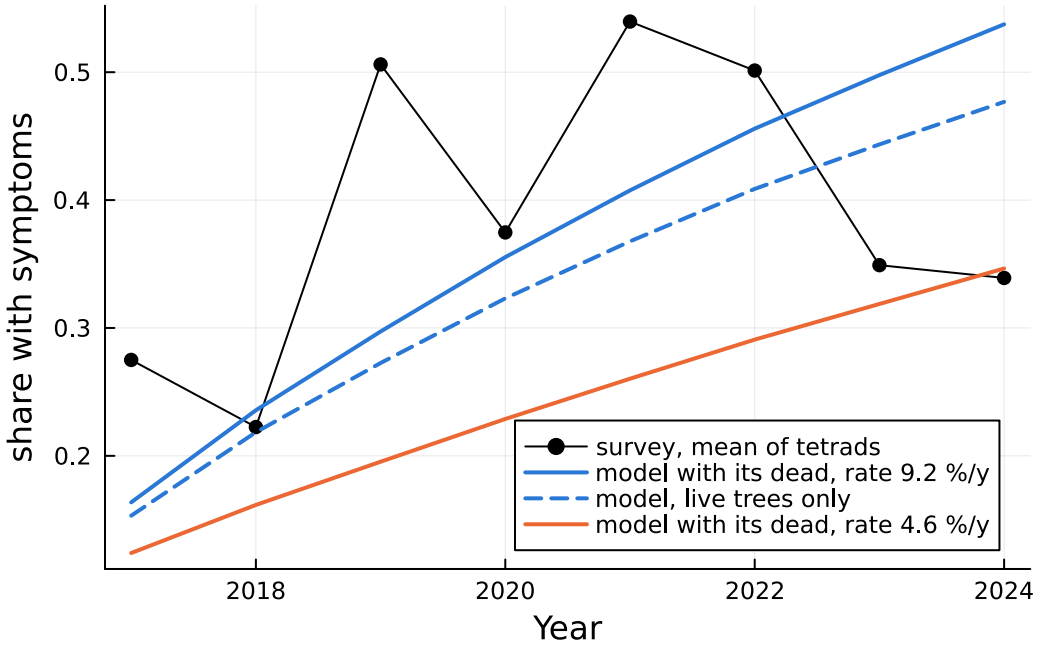
Against Kent's survey

Kent's nine tetrads (2 km squares of ash on or next to public land) give the only long series of crown condition found so far. The survey's "symptoms" appear to include standing dead trees, so the model is shown with and without its own dead. Nothing is fitted. The package holds only the survey's yearly mean over tetrads and the 10 km square each tetrad lies in, placed by hand from its place names:

```
kent_mean = CSV.read(joinpath(datadir(), "gb", "kent_survey_mean.csv"),
 ↪  DataFrame)
kent = CSV.read(joinpath(datadir(), "gb", "kent_squares_10km.csv"),
 ↪  DataFrame)
kent_years = 2017:2024
@assert kent_mean.year == collect(kent_years)
survey_mean = kent_mean.mean_share

function kent_model(rs)
    prob, _ = gb_problem(inputs; B = 15.0, theta = theta_rate(rs),
 ↪  final_year = 2030)
    sol = solve(HostProblem(prob, host_for(0.03, 0.04,
 ↪  SCENARIOS["baseline"])))
    ys = host_years(sol)
    cells = [cell(r.easting_km, r.northing_km) for r in eachrow(kent)]
    shares = map(kent_years) do y
        l = y - first(ys) + 1
        vals = map(cells) do (i, j)
            infected = sum(view(sol.symptomatic, i, j, :, l)) +
 ↪  sum(view(sol.juvenile_symptomatic, i, j, :, l))
            alive = infected + sum(view(sol.healthy, i, j, :, l)) +
 ↪  sum(view(sol.juvenile_healthy, i, j, :, l))
            dead = sol.disease_deaths[i, j, l]  # already cumulative
            (infected / alive, (infected + dead) / (alive + dead))
        end
        (mean(first.(vals)), mean(last.(vals)))
    end
    return first.(shares), last.(shares)
end
central_live, central_dead = kent_model(1.0)
slow_live, slow_dead = kent_model(2.0)
late_years = findall(>=(2019), kent_years)
rms(a) = sqrt(mean((a[late_years] .- survey_mean[late_years]) .^ 2))
plot(kent_years, survey_mean; label = "survey, mean of tetrads", marker =
 ↪  :circle, color = :black, xlabel = "Year", ylabel = "share with
 ↪  symptoms", legend = :bottomright)
plot!(kent_years, central_dead; label = "model with its dead, rate 9.2 %/y",
 ↪  color = BLUE, linewidth = 2)
```

```
plot!(kent_years, central_live; label = "model, live trees only", color =
 ↪  BLUE, linestyle = :dash, linewidth = 2)
plot!(kent_years, slow_dead; label = "model with its dead, rate 4.6 %/y",
 ↪  color = ORANGE, linewidth = 2)
```




```
(; survey_2019_2024 = round(mean(survey_mean[late_years]); digits = 3),
 ↪  model_with_dead = round(mean(central_dead[late_years]); digits = 3),
 rmse_central = round(rms(central_dead); digits = 3), rmse_slow =
 ↪  round(rms(slow_dead); digits = 3),
 survey_2023_24 = round.(survey_mean[end-1:end]; digits = 3), slow_2023_24 =
 ↪  round.(slow_dead[end-1:end]; digits = 3))
```

```
(survey_2019_2024 = 0.435, model_with_dead = 0.425, rmse_central = 0.144,
 ↪  rmse_slow = 0.2, survey_2023_24 = [0.349, 0.339], slow_2023_24 = [0.319,
 ↪  0.346])
```

Over 2019–2024 the survey averages 0.435 and the model with its dead 0.425, with no fitting. But the shape is wrong: the survey rises to about 0.5 by 2019, dips in 2020 and falls after 2022, while the model only rises, as it allows no recovery. The central rate fits the six years better, and the slower rate matches the last two. So q in 2026 stays open at about 0.32 to 0.53. Over that range the 100 % vaccine goes from below s 0.2 planting to above it, and a perfect vaccine from below s 0.05 planting to above it.

The margin over the spore cap

Spore load is capped at S, and at saturation the margin over the cap is $2\beta_0 d$, whatever S is. So the model's weak protection of neighbours comes from that margin, and felling and clearing litter work only if they bring spores below the cap. There are two routes to a smaller margin: lower d through B, which France can't test, and lower β_0, which it can.

So this varies B, at French parameters, and β_0 fixed at 23.26/k with D, S, C_init and κ refitted to France by Nelder–Mead, then run on Great Britain at $B = 25$. The refit is not checked for convergence, so a loss may be overstated:

```
function nelder_mead(f, x0; step = 0.1, maxeval = 300)
    n = length(x0)
    pts = [copy(x0) for _ in 1:(n + 1)]
    foreach(i -> pts[i + 1][i] += step, 1:n)
    vals = f.(pts)
    evals = n + 1
    while evals < maxeval
        o = sortperm(vals)
        pts, vals = pts[o], vals[o]
        c = sum(pts[1:n]) / n
        r = c .+ (c .- pts[end])
        fr = f(r); evals += 1
        if fr < vals[1]
            e = c .+ 2 .* (c .- pts[end])
            fe = f(e); evals += 1
            pts[end], vals[end] = fe < fr ? (e, fe) : (r, fr)
        elseif fr < vals[n]
            pts[end], vals[end] = r, fr
        else
            k = fr < vals[end] ? c .+ 0.5 .* (r .- c) : c .+ 0.5 .*
 ↪  (pts[end] .- c)
            fk = f(k); evals += 1
            if fk < min(fr, vals[end])
                pts[end], vals[end] = k, fk
            else
                for i in 2:(n + 1)
                    pts[i] = pts[1] .+ 0.5 .* (pts[i] .- pts[1])
                    vals[i] = f(pts[i]); evals += 1
                end
            end
        end
    end
    i = argmin(vals)
    return pts[i], vals[i]
end
```

```
function french_refits(ks)
    path = joinpath(RESULTS, "french_refits.csv")
    isfile(path) || write(path, "k,D,S,C_init,kappa,loss\n")
    saved = CSV.read(path, DataFrame)
    todo = [k for k in ks if k ∉ saved.k]
    base = load_peoc_data()
    ref = loglikelihood(AshDiebackProblem(base, [1000.0, 0.0, 23.26, 0.0,
 ↪  18.51, 92.0, 21.50001, 1.0, 0.00855, 0.056]))
    lk = ReentrantLock()
    Threads.@threads :dynamic for n in eachindex(todo)
        k = todo[n]
        ll(x) = begin  # x = log(D, S, C_init, κ)
            v = loglikelihood(AshDiebackProblem(base, [1000.0, 0.0, 23.26 /
 ↪  k, 0.0, exp(x[1]), exp(x[2]), 21.50001, 1.0, exp(x[3]), exp(x[4])]))
            isfinite(v) ? -v : 1e12
        end
        scan = [18.51 * f for f in (1.0, 1.4, 2.0, 2.8, 4.0, 5.6, 8.0, 11.2,
 ↪  16.0, 22.4, 32.0)]
        d0 = scan[argmin([ll(log.([d, 92.0, 0.00855, 0.056])) for d in
 ↪  scan])]
        x, _ = nelder_mead(ll, log.([d0, 92.0, 0.00855, 0.056]); maxeval =
 ↪  250)
        x, v = nelder_mead(ll, x; step = 0.03, maxeval = 150)
        line = join((k, exp(x[1]), exp(x[2]), exp(x[3]), exp(x[4]), v +
 ↪  ref), ",") * "\n"
        lock(() -> open(io -> write(io, line), path, "a"), lk)
    end
    return sort(CSV.read(path, DataFrame), :k)
end
const KS = (1.0, 1.25, 1.5, 2.0, 3.0, 5.0)
refits = french_refits(KS)
const BS = (5.0, 10.0, 15.0, 25.0, 35.0)
data_b = Dict(b => gb_problem(inputs; B = b)[1].data for b in BS)
margin_names = ["baseline", CENTRAL_V, PLANT, "fell 50 %", "clear 50 % of
 ↪  litter", "clear 95 % of litter"]
# k = 5 warns about far-field accuracy at this time step. Only the French
 ↪  likelihood reads those tails.
margin_cases = vcat([case((; setting = "B=$b"), n; data = data_b[b], B = b)
 ↪  for b in BS for n in margin_names],
                    [case((; setting = "k=$(r.k)"), n; data = data25, B =
 ↪  25.0, θ = [1000.0, 0.0, 23.26 / r.k, 0.0, r.D, r.S, 21.50001, 1.0, 0.05,
 ↪  r.kappa])
                     for r in eachrow(refits) for n in margin_names])
margin_runs = sweep("margin", margin_cases)
```

```
mg, mv = lookups(margin_runs).gain, lookups(margin_runs).value
ma, msat = lookups(margin_runs; col = :adults).gain, lookups(margin_runs;
 ↪  col = :saturated).value
nrow(margin_runs)
```

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```

```
md_table(["k", "β₀", "D", "S", "κ", "Log-likelihood lost", "φ for a 95 %
 ↪  cut"],
         [[r.k, fmt(23.26 / r.k, 2), fmt(r.D, 1), fmt(r.S, 1), fmt(r.kappa),
 ↪  fmt(max(r.loss, 0.0), 1), round(Int, r.loss / 1.92)] for r in
 ↪  eachrow(refits)])
margin_table(settings) = md_table(["Gain in healthy adults, 2075",
 ↪  settings...],
    vcat([["Baseline healthy adults", (fmt(mv(s, "baseline")) for s in
     ↪  settings)...]],
         [["Ash at the cap", (fmt(msat(s, "baseline"), 2) for s in
 ↪  settings)...]],
         [[n, (fmt(mg(s, n)) for s in settings)...] for n in
 ↪  margin_names[2:end]],
         [["Fell 50 %, change in all adults", (fmt(ma(s, "fell 50 %")) for s
 ↪  in settings)...]]))
margin_table(["B=$b" for b in BS])
margin_table(["k=$k" for k in KS])
```

k	β_0	D	S	κ	Log-likelihood lost	φ for a 95 % cut
1.0	23.26	18.5	92.0	0.056	0.0	0
1.25	18.61	21.7	92.2	0.057	11.7	6
1.5	15.51	24.1	95.0	0.055	30.2	16
2.0	11.63	29.7	97.4	0.057	83.7	44
3.0	7.75	44.2	97.3	0.065	274.6	143
5.0	4.65	81.5	97.3	0.072	938.6	489

Gain in healthy adults, 2075	B=5.0	B=10.0	B=15.0	B=25.0	B=35.0
Baseline healthy adults	0.107	0.071	0.062	0.055	0.053
Ash at the cap	0.43	0.83	0.93	0.97	0.97

Gain in healthy adults, 2075	B=5.0	B=10.0	B=15.0	B=25.0	B=35.0
central vaccine from 2027	0.077	0.081	0.083	0.083	0.083
plant s 0.2, 2.0 %/y from 2027	0.295	0.324	0.328	0.331	0.333
fell 50 %	0.172	0.083	0.041	0.006	-0.006
clear 50 % of litter	0.124	0.033	0.015	0.007	0.005
clear 95 % of litter	0.399	0.408	0.400	0.347	0.258
Fell 50 %, change in all adults	-0.011	-0.112	-0.156	-0.192	-0.205

Gain in healthy adults, 2075	k=1.0	k=1.25	k=1.5	k=2.0	k=3.0	k=5.0
Baseline healthy adults	0.055	0.057	0.058	0.061	0.073	0.117
Ash at the cap	0.97	0.96	0.94	0.87	0.69	0.04
central vaccine from 2027	0.083	0.083	0.081	0.078	0.076	0.072
plant s 0.2, 2.0 %/y from 2027	0.331	0.330	0.326	0.321	0.312	0.283
fell 50 %	0.006	0.020	0.037	0.069	0.121	0.201
clear 50 % of litter	0.007	0.009	0.013	0.023	0.054	0.164
clear 95 % of litter	0.347	0.382	0.391	0.396	0.399	0.385
Fell 50 %, change in all adults	-0.192	-0.178	-0.159	-0.126	-0.069	0.024

The central vaccine and planting hardly move, because they protect the trees they act on. Felling

and partial litter clearing are what depend on the margin, so B is the input to pin down for those.

Warming

The temperature function is close to a step, so summers past about 21.5 days over 28 °C stop new symptoms in the model. CHESS-SCAPE gives such summers late in the century under RCP8.5. Ash-weighted, for 2071–2080:

```
future = load_gb_future(inputs)
const MEMBERS, GAMMA, KAPPA = ("01", "04", "06", "15"), 21.50001, 0.056
f_T(T) = max(0.0, 1 - T / GAMMA)^KAPPA
w_ash = inputs.ash_ha ./ sum(inputs.ash_ha)
function climate_row(m)
    t = mean(m; dims = 2)
    ft = mean(f_T.(m); dims = 2)
    past = mean(m .>= GAMMA; dims = 2)
    return (sum(w_ash .* t), sum(w_ash .* ft), sum(w_ash .* past))
end
haduk = climate_row(inputs.t28)
climate_range(sc, i) = extrema(climate_row(future[(sc, m)][:, 46:55])[i]
 ↪  for m in MEMBERS)
rng_txt(v, d) = string(fmt(v[1], d), " to ", fmt(v[2], d))
md_table(["", "Mean T28", "Mean f", "Ash past the cutoff"],
         [["HadUK 2004–2025", fmt(haduk[1], 2), fmt(haduk[2]),
 ↪  fmt(haduk[3])],
          ["RCP4.5, four members", rng_txt(climate_range("rcp45", 1), 2),
 ↪  rng_txt(climate_range("rcp45", 2), 2), rng_txt(climate_range("rcp45",
 ↪  3), 2)],
          ["RCP8.5, four members", rng_txt(climate_range("rcp85", 1), 2),
 ↪  rng_txt(climate_range("rcp85", 2), 2), rng_txt(climate_range("rcp85",
 ↪  3), 2)]])
```

	Mean T28	Mean f	Ash past the cutoff
HadUK 2004–2025	2.42	0.992	0.000
RCP4.5, four members	3.98 to 10.12	0.81 to 0.98	0.00 to 0.16
RCP8.5, four members	12.06 to 25.43	0.36 to 0.79	0.19 to 0.62

Yet the effect on results by 2075 is small:

```
climates = vcat([("none", "–")], [(sc, m) for sc in ("rcp45", "rcp85") for m
 ↪  in MEMBERS])
data_for(sc, m) = sc == "none" ? DATA15 : gb_problem(inputs; B = 15.0,
 ↪  temperature = future[(sc, m)])[1].data
```

```
warm_names = ["baseline", "baseline s 0.05", "baseline s 0.5", CENTRAL_V,
 ↪  HUNDRED_V, PERFECT_V, PLANT, PLANT_T, PLANT_W, PLANT_O, "fell 50 %",
 ↪  "clear 95 % of litter"]
warm_cases = reduce(vcat, [(d = data_for(sc, m); [case((; climate = sc,
 ↪  member = m), n; data = d) for n in warm_names]) for (sc, m) in
 ↪  climates])
warm_runs = sweep("warming", warm_cases)
wg, wv = lookups(warm_runs).gain, lookups(warm_runs).value
ws, wn = lookups(warm_runs; col = :healthy_south).value, lookups(warm_runs;
 ↪  col = :healthy_north).value
nrow(warm_runs)
```

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```

```
wrange(sc, f) = extrema(f(sc, m) for m in MEMBERS)
txt(v) = string(fmt(v[1]), "–", fmt(v[2]))
md_table(["Healthy adults, 2075", "No warming", "RCP4.5", "RCP8.5"],
         [["Baseline, all GB", fmt(wv("none", "–", "baseline")),
 ↪  txt(wrange("rcp45", (s, m) -> wv(s, m, "baseline"))),
 ↪  txt(wrange("rcp85", (s, m) -> wv(s, m, "baseline")))],
          ["Baseline, below 300 km north", fmt(ws("none", "–", "baseline")),
 ↪  txt(wrange("rcp45", (s, m) -> ws(s, m, "baseline"))),
 ↪  txt(wrange("rcp85", (s, m) -> ws(s, m, "baseline")))],
          ["Baseline, from 600 km north", fmt(wn("none", "–", "baseline")),
 ↪  txt(wrange("rcp45", (s, m) -> wn(s, m, "baseline"))),
 ↪  txt(wrange("rcp85", (s, m) -> wn(s, m, "baseline")))]])
md_table(["Gain in healthy adults, 2075", "No warming", "RCP4.5", "RCP8.5"],
         [[n, fmt(wg("none", "–", n)), txt(wrange("rcp45", (s, m) -> wg(s,
 ↪  m, n))), txt(wrange("rcp85", (s, m) -> wg(s, m, n)))]
          for n in (PLANT_T, PERFECT_V, "clear 95 % of litter", PLANT,
 ↪  HUNDRED_V, PLANT_W, CENTRAL_V, PLANT_O, "fell 50 %")])
```

Healthy adults, 2075	No warming	RCP4.5	RCP8.5
Baseline, all GB	0.062	0.062–0.068	0.066–0.090
Baseline, below 300 km north	0.051	0.052–0.060	0.058–0.091
Baseline, from 600 km north	0.188	0.188–0.188	0.188–0.188

Gain in healthy adults, 2075	No warming	RCP4.5	RCP8.5
plant s 0.05, 2.0 %/y from 2027	0.477	0.477–0.478	0.477–0.479

Gain in healthy adults, 2075	No warming	RCP4.5	RCP8.5
perfect vaccine from 2027	0.448	0.444–0.448	0.432–0.445
clear 95 % of litter	0.400	0.396–0.400	0.380–0.396
plant s 0.2, 2.0 %/y from 2027	0.328	0.329–0.337	0.335–0.357
100 % vaccine from 2027	0.246	0.246–0.250	0.248–0.258
plant s 0.5, 2.0 %/y from 2027	0.165	0.165–0.177	0.174–0.213
central vaccine from 2027	0.083	0.083–0.085	0.084–0.092
plant ordinary, 2.0 %/y from 2027	0.072	0.073–0.083	0.080–0.127
fell 50 %	0.041	0.036–0.041	0.017–0.038

The likely reason, not tested separately here, is that the hot decades come late, after most trees have symptoms, and warming acts only on new symptoms, which don't reverse. This is an upper limit on what warming does for ash, given the rest of the model, since spores, litter, death and ash growth don't respond to it. Drought and heat stress on ash itself aren't in the model at all. RCP4.5 is scaled from RCP8.5, and the scenarios stop in 2080.

The French parameters

Vignette 5 shows that the French survey counts are about 20 times as variable as binomial, and gives two sets of draws: 20 from the published posterior ($\varphi = 1$) and 60 allowing for that ($\varphi = 20$). Here each draw runs under every scenario, at B = 15 and 25, so gains and ranks are paired within a draw. Only β_0, D, S and κ come from France. C_init is France's start, so here it stays at 0.05 in the seed squares:

```
draws = CSV.read(joinpath(@__DIR__, "..", "05_reproducing_the_paper",
 ↪  "results", "posterior_draws.csv"), DataFrame; stringtype = String)
post_names = ["baseline", "baseline s 0.05", "baseline s 0.5", ranked...]
post_cases = [case((; set = r.set, draw = string(r.draw), B =
 ↪  string(Int(b)), beta_0 = @sprintf("%.6f", r.beta_0)), n;
                   data = b == 15.0 ? DATA15 : data25, B = b,
                   θ = [1000.0, 0.0, r.beta_0, 0.0, r.D, r.S_sat, 21.50001,
 ↪  1.0, 0.05, r.kappa])
              for b in (15.0, 25.0) for r in eachrow(draws) for n in
 ↪  post_names]
post_runs = sweep("posterior", post_cases)
draw_table = [(r.set, r.draw, @sprintf("%.6f", r.beta_0)) for r in
 ↪  eachrow(draws)]
```

```
pl75, pl50, pa75 = lookups(post_runs), lookups(post_runs; year = 2050),
 ↪  lookups(post_runs; col = :adults)
over(g, set, b, n) = [g(s, d, b, β, n) for (s, d, β) in draw_table if s ==
 ↪  set]
nrow(post_runs)
```

```
6720
```

The ranking, with 95 % of the $\varphi = 20$ draws, at B = 15:

```
md_table(["Intervention, from 2027", "2050", "2075", "Published posterior,
 ↪  2075", "Change in all adults, 2075"],
         [[n, fmti(over(pl50.gain, "phi20", 15, n)), "**" *
 ↪  fmti(over(pl75.gain, "phi20", 15, n)) * "**", fmti(over(pl75.gain,
 ↪  "phi1", 15, n)),
           @sprintf("%+.2f", mean(over(pa75.gain, "phi20", 15, n)))] for n
            ↪  in ranked])
```

Intervention, from 2027	2050	2075	Published posterior, 2075	Change in all adults, 2075
plant s 0.05, 2.0 %/y from 2027	0.197 [0.195, 0.199]	0.478 [0.472, 0.483]	0.477 [0.476, 0.479]	+0.52
perfect vaccine from 2027	0.318 [0.305, 0.332]	0.448 [0.430, 0.468]	0.448 [0.445, 0.451]	+0.33
clear 95 % of litter	0.226 [0.133, 0.265]	0.374 [0.279, 0.414]	0.399 [0.394, 0.402]	+0.25
central vaccine + planting	0.247 [0.237, 0.260]	0.364 [0.351, 0.377]	0.364 [0.362, 0.365]	+0.53
plant s 0.2, 2.0 %/y from 2027	0.161 [0.157, 0.165]	0.328 [0.317, 0.339]	0.328 [0.326, 0.329]	+0.51
100 % vaccine from 2027	0.210 [0.197, 0.225]	0.244 [0.226, 0.266]	0.245 [0.242, 0.248]	+0.20
plant s 0.5, 2.0 %/y from 2027	0.107 [0.102, 0.114]	0.164 [0.152, 0.177]	0.164 [0.162, 0.166]	+0.50
central vaccine from 2027	0.091 [0.083, 0.099]	0.082 [0.076, 0.089]	0.083 [0.081, 0.084]	+0.08
plant ordinary, 2.0 %/y from 2027	0.057 [0.052, 0.063]	0.071 [0.063, 0.082]	0.072 [0.070, 0.073]	+0.25
fell 50 %	0.005 [-0.006, 0.030]	0.034 [-0.006, 0.092]	0.042 [0.034, 0.050]	-0.16
clear 50 % of litter	0.006 [0.003, 0.012]	0.014 [0.005, 0.035]	0.015 [0.013, 0.017]	+0.01

At B = 25, and how often each pair keeps its order, over the same draws:

```
share_beating(set, b, x, y) = mean(over(pl75.gain, set, b, x) .>
 ↪  over(pl75.gain, set, b, y))
pairs_p = ((PLANT_T, PERFECT_V), (PERFECT_V, "clear 95 % of litter"),
 ↪  ("clear 95 % of litter", PLANT), (PLANT, HUNDRED_V), (PLANT, CENTRAL_V),
          (PLANT_W, CENTRAL_V), (CENTRAL_V, PLANT_O), (CENTRAL_V, "fell 50
 ↪  %"), ("fell 50 %", PLANT_O), ("fell 50 %", "clear 50 % of litter"))
md_table(["First beats second, share of draws", "φ = 1, B 15", "φ = 20, B
 ↪  15", "φ = 1, B 25", "φ = 20, B 25"],
         [[string(x, " > ", y), (fmt(share_beating(s, b, x, y), 2) for b in
 ↪  (15, 25) for s in ("phi1", "phi20"))...] for (x, y) in pairs_p])
md_table(["Near a threshold, φ = 20", "B = 15", "B = 25"],
         [[n, fmti(over(pl75.gain, "phi20", 15, n)), fmti(over(pl75.gain,
 ↪  "phi20", 25, n))] for n in ("clear 95 % of litter", "fell 50 %", "clear
 ↪  50 % of litter")])
```

First beats second, share of draws	φ = 1, B 15	φ = 20, B 15	φ = 1, B 25	φ = 20, B 25
plant s 0.05, 2.0 %/y from 2027 > perfect vaccine from 2027	1.00	1.00	1.00	1.00
perfect vaccine from 2027 > clear 95 % of litter	1.00	1.00	1.00	1.00
clear 95 % of litter > plant s 0.2, 2.0 %/y from 2027	1.00	0.83	0.85	0.45
plant s 0.2, 2.0 %/y from 2027 > 100 % vaccine from 2027	1.00	1.00	1.00	1.00
plant s 0.2, 2.0 %/y from 2027 > central vaccine from 2027	1.00	1.00	1.00	1.00
plant s 0.5, 2.0 %/y from 2027 > central vaccine from 2027	1.00	1.00	1.00	1.00

First beats second, share of draws	φ = 1, B 15	φ = 20, B 15	φ = 1, B 25	φ = 20, B 25
central vaccine from 2027 > plant ordinary, 2.0 %/y from 2027	1.00	1.00	1.00	1.00
central vaccine from 2027 > fell 50 %	1.00	0.93	1.00	1.00
fell 50 % > plant ordinary, 2.0 %/y from 2027	0.00	0.08	0.00	0.00
fell 50 % > clear 50 % of litter	1.00	0.77	0.45	0.38

Near a threshold, φ = 20	B = 15	B = 25
clear 95 % of litter	0.374 [0.279, 0.414]	0.296 [0.137, 0.390]
fell 50 %	0.034 [-0.006, 0.092]	0.005 [-0.016, 0.041]
clear 50 % of litter	0.014 [0.005, 0.035]	0.006 [0.002, 0.013]

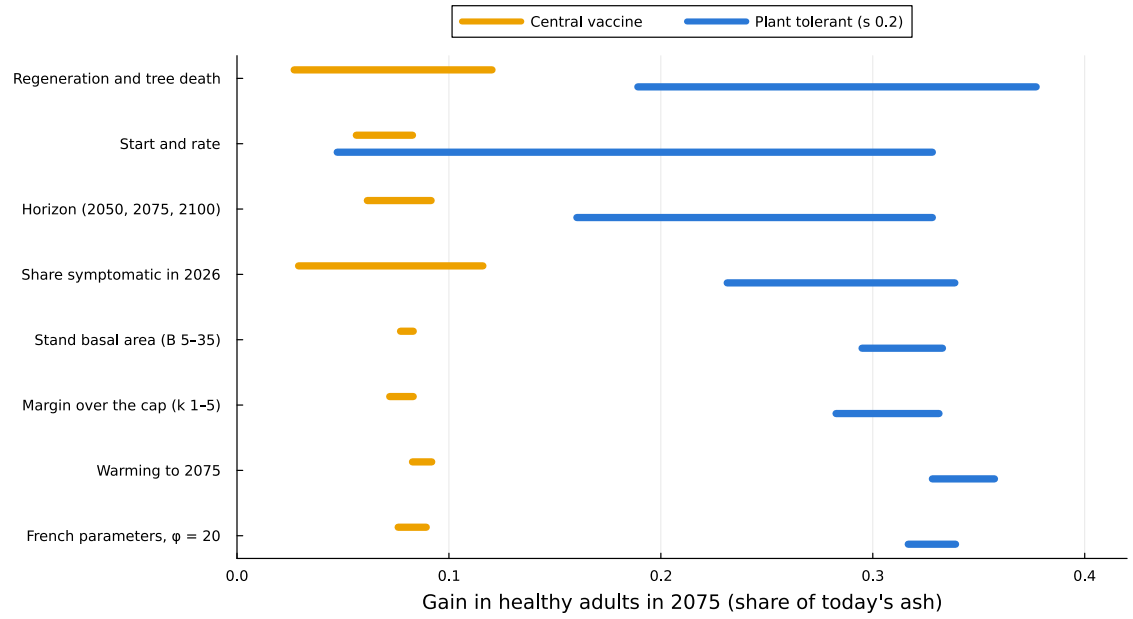
Vaccine and planting gains barely move, even allowing for φ = 20. Measures near a threshold do, because they work only if they bring spores below the cap, which depends on β_0 and D.

What changes the result

Pulling these together, for the central vaccine and tolerant planting, each unknown one at a time:

```
rows_m = [("Regeneration and tree death", extrema(gg(ρ, μ, CENTRAL_V) for
 ↪  (ρ, μ) in corners), extrema(gg(ρ, μ, PLANT) for (ρ, μ) in corners)),
          ("Start and rate", extrema(gain_y(n, 2075) for n in (CENTRAL_V,
 ↪  "central vaccine from 2040")),
           extrema(gain_y(plant_name(0.2, r, s), 2075) for r in (0.005,
            ↪  0.01, 0.02) for s in (2027, 2035, 2040, 2050))),
          ("Horizon (2050, 2075, 2100)", extrema(gain_y(CENTRAL_V, y) for y
 ↪  in (2050, 2075, 2100)), extrema(gain_y(PLANT, y) for y in (2050, 2075,
 ↪  2100))),
          ("Share symptomatic in 2026", extrema(sg(l, CENTRAL_V) for (l, _,
 ↪  _) in share_settings), extrema(sg(l, PLANT) for (l, _, _) in
 ↪  share_settings)),
```

```
        ("Stand basal area (B 5–35)", extrema(mg("B=$b", CENTRAL_V) for b
 ↪  in BS), extrema(mg("B=$b", PLANT) for b in BS)),
        ("Margin over the cap (k 1–5)", extrema(mg("k=$k", CENTRAL_V) for
 ↪  k in KS), extrema(mg("k=$k", PLANT) for k in KS)),
        ("Warming to 2075", extrema(wg(s, m, CENTRAL_V) for (s, m) in
 ↪  climates), extrema(wg(s, m, PLANT) for (s, m) in climates)),
        ("French parameters, φ = 20", interval(over(pl75.gain, "phi20",
 ↪  15, CENTRAL_V)), interval(over(pl75.gain, "phi20", 15, PLANT)))]
fig_m = plot(; yticks = (1:length(rows_m), reverse(first.(rows_m))), xlims
 ↪  = (0, 0.42), xlabel = "Gain in healthy adults in 2075 (share of today's
 ↪  ash)",
           legend = :outertop, legendcolumns = 2, size = (950, 520),
 ↪  left_margin = 4Plots.mm, bottom_margin = 4Plots.mm, grid = :x)
for (i, r) in enumerate(reverse(rows_m))
    plot!(fig_m, collect(r[2]), [i + 0.13, i + 0.13]; color = YELLOW,
     ↪  linewidth = 6, label = i == 1 ? "Central vaccine" : "")
    plot!(fig_m, collect(r[3]), [i - 0.13, i - 0.13]; color = BLUE,
     ↪  linewidth = 6, label = i == 1 ? "Plant tolerant (s 0.2)" : "")
end
fig_m
```




```
md_table(["Unknown", "Central vaccine", "Tolerant planting (s 0.2)"],
 ↪  [[r[1], txt(r[2]), txt(r[3])] for r in rows_m])
```

Unknown	Central vaccine	Tolerant planting (s 0.2)
Regeneration and tree death	0.027–0.120	0.189–0.377
Start and rate	0.056–0.083	0.047–0.328
Horizon (2050, 2075, 2100)	0.062–0.092	0.160–0.328
Share symptomatic in 2026	0.029–0.116	0.231–0.339
Stand basal area (B 5–35)	0.077–0.083	0.295–0.333
Margin over the cap (k 1–5)	0.072–0.083	0.283–0.331
Warming to 2075	0.083–0.092	0.328–0.357
French parameters, $\varphi = 20$	0.076–0.089	0.317–0.339

Quality of the product, in the section above, moves them as far again, and it is set by the input chosen. Regeneration and tree death, the share symptomatic and the margin use B = 25, and the others use B = 15. Those two values of B give gains within 0.003 for these two interventions. Felling and litter are not shown, as they depend on B and the margin.

Summary

- Planting tolerant stock at scale gives the largest gains by 2075, if it can start soon. Planting 2 % a year from 2027 gains 0.33 of today's ash as healthy adults at the central tolerance and 0.48 at s 0.05, against 0.08 for the central vaccine, but about 5000 ha of ash a year is probably not achievable.
- Slower or later planting loses that lead. At 0.5 % a year from 2027, planting passes the central vaccine started in 2027 only in 2071. From 2035 at that rate, it does not pass it by 2075.
- Vaccines pay sooner. If both start in 2027, a 100 % vaccine leads planting at the central tolerance until 2058, and a perfect vaccine leads very tolerant planting until 2068.
- A vaccine's value lies in efficacy, duration and reach, and falls with delay. The central vaccine gains 0.08. Efficacy of 0.5 gives 0.03, protection lasting 5 years 0.03, and a start in 2040 0.06.
- Regeneration and the share already symptomatic can change the order. The 100 % vaccine beats planting at the central tolerance in 4 of the 9 cells of the regeneration grid, and when 0.32 of ash is symptomatic in 2026 rather than 0.53.
- Felling and clearing part of the litter do little for healthy trees. Felling half of symptomatic adults a year gains 0.03 healthy adults and changes all adults by -0.16. Clearing 50 % of litter gains 0.01. At the best-fit parameters, clearing 95 % ends the epidemic in the model, but it means treating about 160000 ha a year, and its effect is the most uncertain here (0.14 to 0.41 over B and the French parameters).

What field data would help most

These are ordered by how much they could change the ranking above.

1. Field performance and supply of tolerant stock. How much less often do selected trees develop symptoms and die in the field, and how many could be planted a year, from when?
2. Long-term regeneration. It decides the long-run result and whether a 100 % vaccine beats planting. Repeat counts of seedlings and saplings in affected stands would address it.
3. Repeated surveys of symptoms and deaths with a fixed protocol, including recovery. Kent's fall after 2022 could be recovery, felling or a change of observers. If it is recovery, it challenges "no recovery", which underlies the model.
4. For a vaccine: efficacy, duration and how fast it could reach trees. All are assumed here, and they matter more than the coverage cap.
5. Ash basal area including trees outside woodland. This matters for felling and partial litter clearing only.
6. A test below the spore cap. The French data say little about how the disease behaves when inoculum falls below the cap, which is where litter clearing works or fails.

Limits

- Nothing is fitted to Britain, and the check against Kent uses nine small tetrads on public land, matching in level and not in shape.
- Inputs without evidence: every vaccine input, tolerant stock's traits, the native tolerant share and regeneration. These are sweeps, not estimates.
- The model has no recovery, and no stand scale. Spores mix across each 10 km square, so a stand protected by clearing its own litter can't appear here, and such benefits may be larger than this suggests.
- No costs. Areas treated are not costs, and a hectare planted differs from a hectare vaccinated. Felling's reasons of safety and amenity are outside the measure.
- Few draws, one change at a time. The outer limits of each interval are rough. Apart from the French parameters with B, the unknowns are varied one at a time, so joint extremes aren't tested.
- Planting availability is unchecked. Current rules on moving or planting ash, and nursery capacity, aren't established here.